

Education Readiness Risk Modeling in R

Purpose of the Study

This project asks a practical planning question: when support capacity is limited, which public-safe assessment transitions should be reviewed first before the next assessment window?

The analysis turns current readiness, attendance, assessment-window timing, and course context into a probability of next-window support risk. The goal is not to label a student or automate an academic decision. The goal is to create a transparent review queue that helps a support team focus attention, understand the tradeoffs, and monitor whether the process is working.

Recommendation

Use the model as a **human review prioritization tool**. A 50% support-review threshold is a reasonable starting point for planning because it balances coverage and workload: it flags 326 of 666 holdout transitions (48.9%) and captures 298 of 347 observed support-risk cases.

If capacity is tighter, the 65% threshold is the next practical option: it flags 283 transitions and captures 267 of 347 observed support-risk cases. The final threshold should be set from available review capacity, intervention cost, and tolerance for missed support needs.

What This Means Operationally

Decision	Practical answer
Who should be reviewed first?	Begin with transitions above the support-review threshold, then use educator context before taking action.
What workload does the starting threshold create?	A 50% threshold flags 326 of 666 holdout transitions (48.9%).
What coverage does the starting threshold provide?	That threshold captures 298 of 347 observed support-risk cases, or 85.9% of the cases in the holdout set.
What if review capacity is tighter?	A 65% threshold flags 283 transitions and captures 267 of 347 observed support-risk cases.
What should the model not do?	It should not automatically assign intervention, placement, grading, or discipline decisions.

The score should be used to decide what gets reviewed first, not what happens automatically. A support team would still confirm context, look at recent trajectory, and decide whether any action is appropriate.

Key Findings in Plain English

1. Current readiness is the strongest planning signal. Context-only models were much weaker, which means the assessment-readiness evidence adds real prioritization value.
2. The readiness relationship is not just a straight line. Risk changes sharply across readiness regions, so the final model keeps a threshold-like shape while staying explainable.
3. A risk threshold is a staffing decision. Lower thresholds review more transitions and miss fewer support-risk cases; higher thresholds focus effort but leave more cases outside the queue.
4. Risk categories are most useful as workflow labels. They translate probabilities into monitoring, watch-list, review, and priority-review actions.

The ranking view is useful even before choosing a hard cutoff: the highest-risk decile has 1.92x lift over the base rate, and the top two deciles capture 38.3% of observed support-risk cases.

Risk Categories and Suggested Actions

Risk categories make the model easier to use in a planning conversation. They are operating labels for a public-safe portfolio analysis, not permanent labels for real students.

Category	Transitions	Observed risk	Observed cases	Suggested use
Monitor	298	11.1%	33	Routine monitoring; no added review solely from this score.
Watch	42	38.1%	16	Check trend and attendance context before the next assessment.
Review	43	72.1%	31	Add to the support-team review queue.
Priority	283	94.3%	267	Review first when support capacity is limited.

Data Used

The analysis uses a public-safe assessment extract with one row per assessment window. The modeling table turns consecutive assessment windows into prediction records: current assessment information is used to predict support risk at the next assessment.

Measure	Value
Raw assessment rows	4,018
Modeled transitions	3,322
Unique public-safe student IDs	696
Support-risk event rate	52.1%
Current nonparticipation rate	7.2%
Next-window nonparticipation rate	7.2%
Median current readiness	47.7
Included assessment windows	beginning-of-year, end-of-year

The extract uses simulated identifiers and generalized assessment behavior from a bootstrapped assessment workflow. It should not be treated as a release of real student-level records.

Technical Validation Summary

The selected technical model is **Piecewise readiness**. On the holdout set, it achieved AUC **0.938**, log loss **0.309**, and Brier score **0.093**. Repeated cross-validation produced log loss **0.318** and AUC **0.935**.

The holdout event rate is 52.1%, so the evaluation is not based on a rare-event edge case. Bootstrap intervals, calibration diagnostics, lift checks, subgroup calibration, and sensitivity testing are included below.

Direct Answers

1. The primary modeled outcome is next-window support risk, defined as a next assessment score below 50 or next-window nonparticipation. The holdout event rate is 52.1%.
2. The best operating model is **Piecewise readiness**, selected from interpretable GLM candidates after comparing nonlinear and benchmark model families.

3. The mathematical discovery step matters: Nonparametric smoothing supported a threshold-like readiness curve; piecewise and polynomial candidates were tested against spline and periodic benchmarks.
4. The model is strongest as a prioritization tool. Thresholds convert probabilities into workload, missed-risk, and precision tradeoffs.
5. The analysis is public-safe: it uses simulated identifiers and generalized assessment behavior, and excludes private prompts, exams, real student-identifiable records, credentials, and private source documents.

Technical Appendix: Model Journey

The model search follows a disciplined workflow: inspect the shape first, test candidate parametric families second, and keep the final model interpretable unless a flexible benchmark clearly earns its complexity.

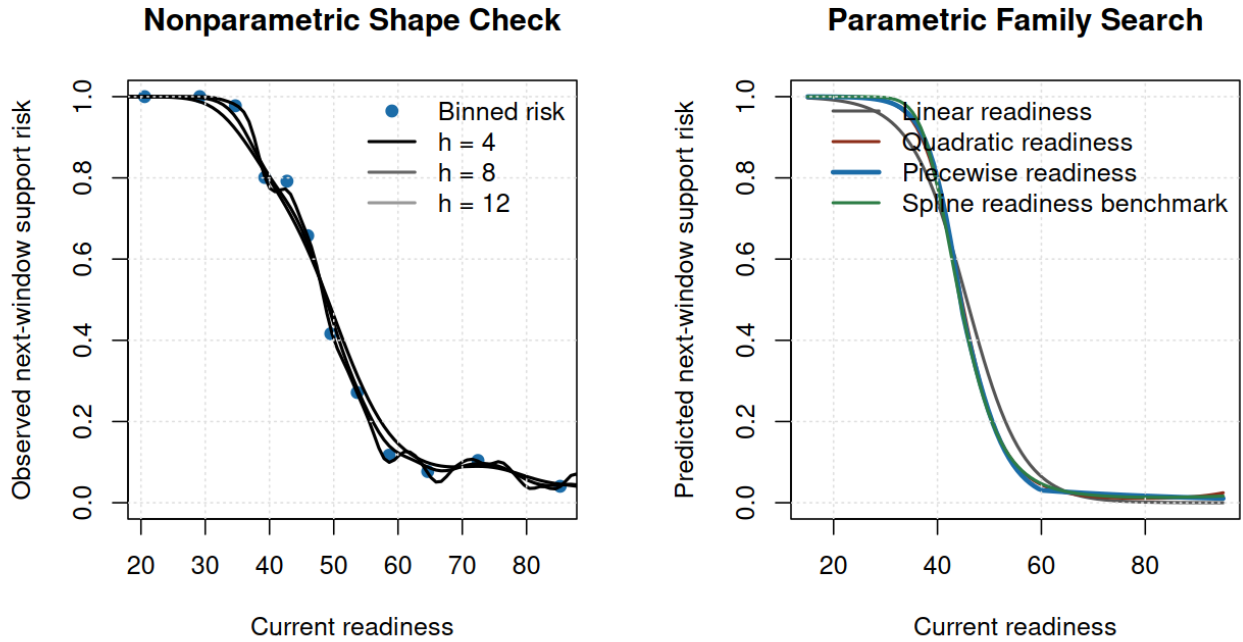


Figure 1: Nonparametric and parametric shape discovery

Family	Why tested	Decision	CV loss	Holdout AUC
Context baseline	Tests whether demographic and operating context alone is enough.	Rejected; validation is much weaker without readiness.	0.643	0.673
Linear readiness	Adds the main readiness signal with a simple monotone probability shape.	Rejected; ranking is strong but probability quality is worse.	0.340	0.933
Quadratic readiness	Tests whether risk accelerates near low readiness values.	Rejected; close to selected model, but less directly aligned with the discovered threshold shape.	0.319	0.937

Family	Why tested	Decision	CV loss	Holdout AUC
Cubic polynomial readiness	Checks whether a more flexible polynomial improves fit enough to justify instability risk.	Rejected; added polynomial curvature without improving the operating story.	0.318	0.936
Piecewise readiness	Uses the smooth shape discovery to encode separate readiness regions.	Selected as the operating model.	0.318	0.938
Periodic context benchmark	Tests recurring assessment-window structure without making periodicity the headline.	Benchmark only; periodic terms did not justify replacing the operating model.	0.320	0.937
Spline readiness benchmark	Flexible nonlinear benchmark for the readiness curve.	Benchmark only; used to test whether flexible curvature changes the conclusion.	0.318	0.937

Candidate logistic models were then compared with repeated stratified 5-fold cross-validation on the training split. Log loss is the primary criterion because a support-prioritization workflow needs useful probabilities, not only rank ordering.

Model	Selected	Role	Params	CV loss	CV SD	CV AUC	Holdout loss	Holdout AUC	Delta
Piecewise readiness	Yes	Selection candidate	15	0.318	0.001	0.935	0.309	0.938	0.000
Cubic polynomial readiness		Selection candidate	15	0.318	0.001	0.935	0.313	0.936	0.000
Spline readiness benchmark		Benchmark	16	0.318	0.001	0.935	0.313	0.937	0.000
Quadratic readiness		Selection candidate	14	0.319	0.001	0.935	0.316	0.937	0.001
Periodic context benchmark		Benchmark	16	0.320	0.001	0.935	0.315	0.937	0.002
Linear readiness		Selection candidate	13	0.340	0.001	0.935	0.358	0.933	0.022
Context baseline		Selection candidate	10	0.643	0.000	0.674	0.644	0.673	0.325

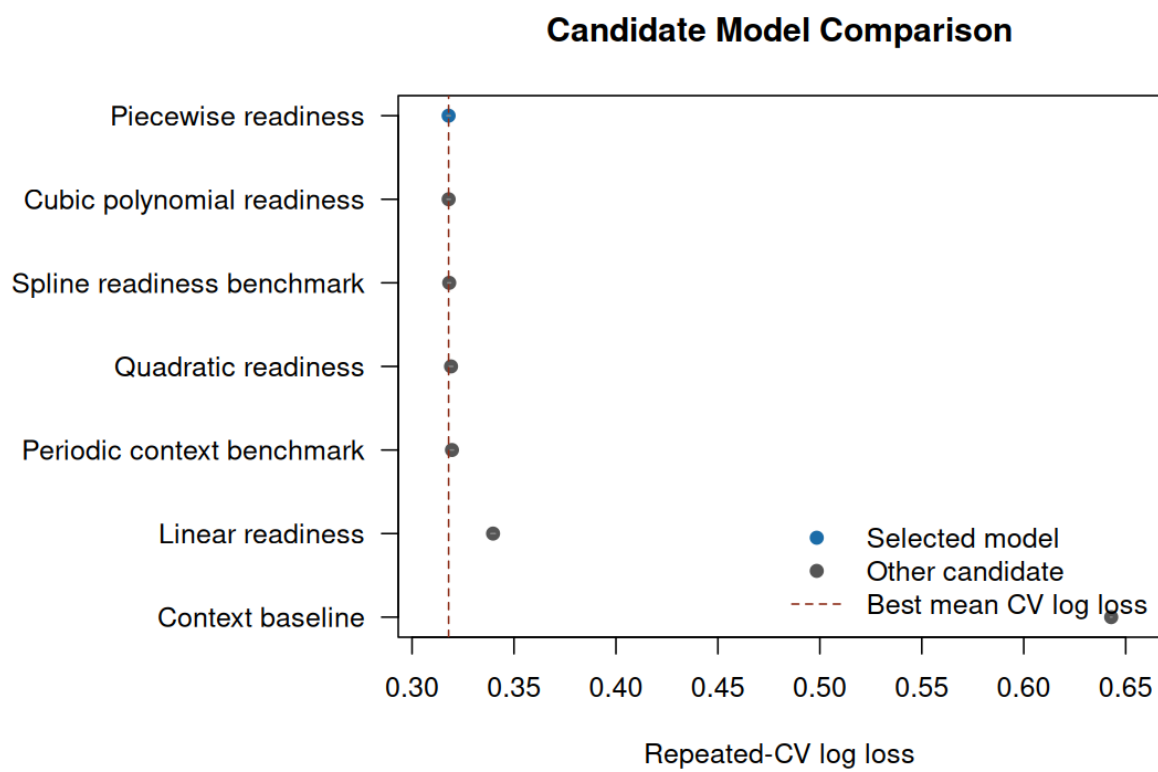


Figure 2: Candidate model comparison

The selection rule favors the simplest non-benchmark model within one standard error of the best repeated-CV log loss. That rule protects the portfolio story from choosing a visually impressive model that does not materially improve validated probability quality.

Technical Appendix: Final Model

Metric	Value
Selected model	Piecewise readiness
Selection rule	Simplest non-benchmark model within one standard error of best repeated-CV log loss
Shape discovery result	Nonparametric smoothing supported a threshold-like readiness curve; piecewise and polynomial candidates were tested against spline and periodic benchmarks.
Training rows	2656
Holdout rows	666
Training event rate	52.1%
Holdout event rate	52.1%
Repeated CV folds	5
Repeated CV repeats	6
CV log loss	0.318
CV AUC	0.935
Holdout log loss	0.309
Holdout Brier score	0.093
Holdout AUC	0.938
Calibration intercept	0.165
Calibration slope	1.025
Training log loss	0.312
Training AUC	0.938

Bootstrap intervals give a practical uncertainty band around the holdout metrics.

Metric	Estimate	95% CI
LogLoss	0.309	0.268 to 0.352
Brier	0.093	0.079 to 0.107
AUC	0.938	0.919 to 0.955

The adjusted odds ratios below translate the selected GLM into stakeholder-readable effects.

Predictor	Scale	Odds ratio	95% CI	p-value
Grade 10 vs grade 9	1-unit / level change	1.03	0.74 to 1.44	0.869
Grade 11 vs grade 9	1-unit / level change	1.11	0.78 to 1.57	0.576
Grade 12 vs grade 9	1-unit / level change	1.08	0.65 to 1.80	0.770
Honors track vs regular	1-unit / level change	1.27	0.85 to 1.91	0.242
AP track vs regular	1-unit / level change	1.25	0.91 to 1.73	0.173
Beyond-core track vs regular	1-unit / level change	2.78	0.69 to 11.29	0.152
Current window: end of year	1-unit / level change	5.62	4.18 to 7.56	<0.001
Attendance category: high absence vs normal	1-unit / level change	0.83	0.57 to 1.23	0.363

Predictor	Scale	Odds ratio	95% CI	p-value
Attendance category: at-risk absence vs normal	1-unit / level change	1.26	0.41 to 3.88	0.689
Attendance probability	0.1-unit change	0.80	0.50 to 1.29	0.366
Readiness shortfall below 45	5-unit change	4.73	3.60 to 6.23	<0.001
Readiness gain from 45 to 60	5-unit change	0.34	0.29 to 0.39	<0.001
Readiness gain above 60	5-unit change	0.86	0.75 to 0.98	0.023
School-year sequence	1-unit / level change	0.97	0.91 to 1.04	0.378

Technical Appendix: Probability Scale

Risk categories provide a bridge between calibrated probabilities and support workflows.

Category	Transitions	Share	Pred	Obs	Cases
Monitor	298	44.7%	11.0%	11.1%	33
Watch	42	6.3%	41.1%	38.1%	16
Review	43	6.5%	56.4%	72.1%	31
Priority	283	42.5%	92.5%	94.3%	267

A threshold turns probabilities into a work queue. Lower thresholds catch more support-risk cases but create more reviews; higher thresholds focus capacity but miss more students.

Threshold	Flagged	Flagged %	Captured	Sens	Spec	PPV	NPV
35%	368	55.3%	314 of 347	90.5%	83.1%	85.3%	88.9%
45%	336	50.5%	304 of 347	87.6%	90.0%	90.5%	87.0%
50%	326	48.9%	298 of 347	85.9%	91.2%	91.4%	85.6%
55%	310	46.5%	286 of 347	82.4%	92.5%	92.3%	82.9%
65%	283	42.5%	267 of 347	76.9%	95.0%	94.3%	79.1%
75%	259	38.9%	250 of 347	72.0%	97.2%	96.5%	76.2%

The table below uses illustrative support-planning economics to show how a threshold can be chosen from capacity and intervention assumptions. These values are scenario assumptions, not claims about a real school system.

Threshold	Flagged	Captured	Benefit	Cost	Net
35%	368	314	\$70,650	\$27,600	\$43,050
45%	336	304	\$68,400	\$25,200	\$43,200
50%	326	298	\$67,050	\$24,450	\$42,600
55%	310	286	\$64,350	\$23,250	\$41,100
65%	283	267	\$60,075	\$21,225	\$38,850
75%	259	250	\$56,250	\$19,425	\$36,825

Technical Appendix: Model Checks

ROC checks ranking quality. Calibration checks whether predicted probabilities are on the right scale across ordered risk bands.

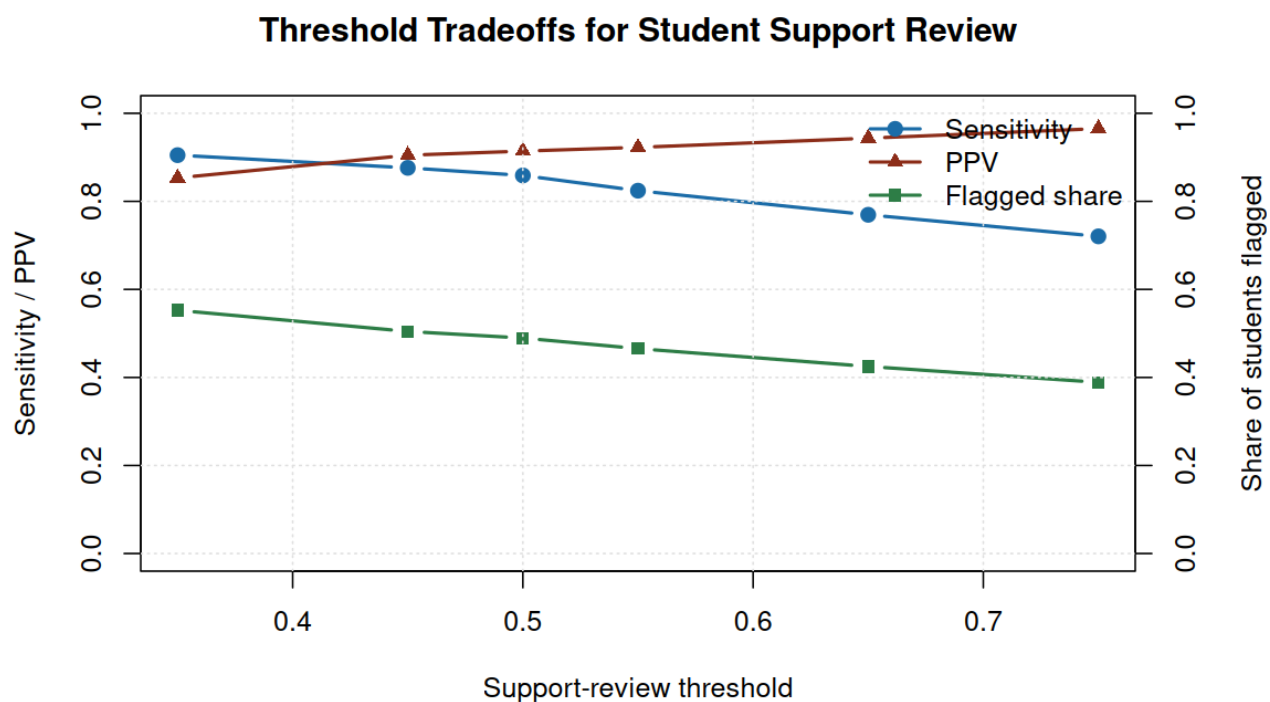


Figure 3: Threshold tradeoffs

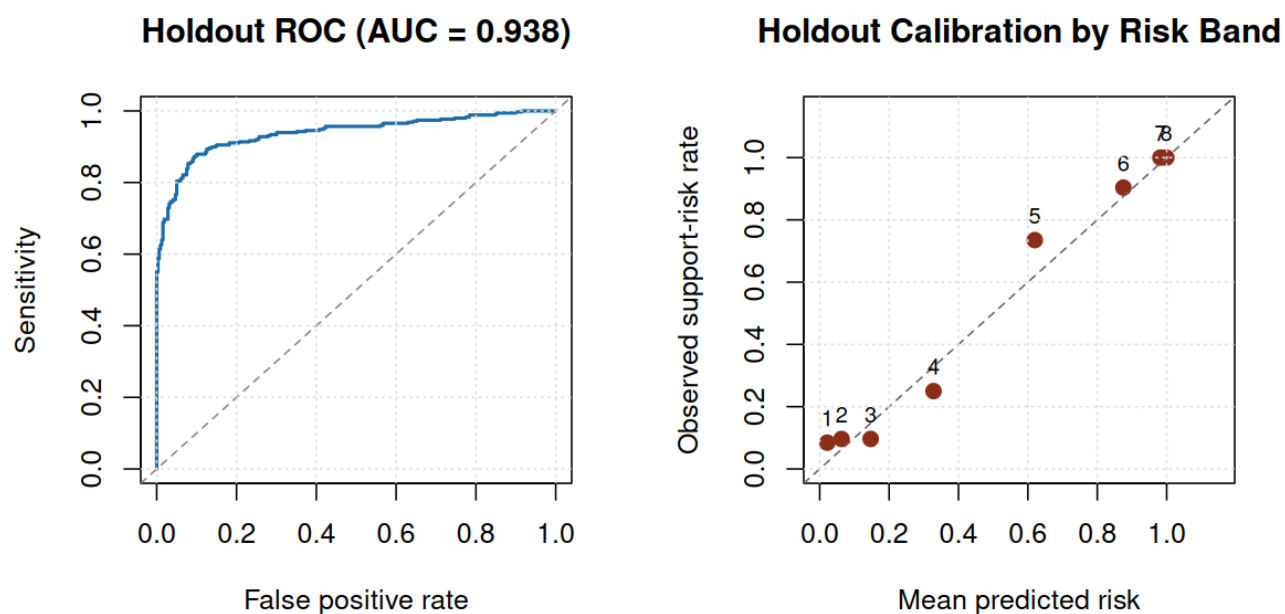


Figure 4: ROC and calibration diagnostics

Band	N	Pred	Obs	Expected	Cases
Band 1	83	2.2%	8.4%	1.8	7
Band 2	83	6.3%	9.6%	5.2	8
Band 3	83	14.7%	9.6%	12.2	8
Band 4	84	32.8%	25.0%	27.6	21
Band 5	83	62.0%	73.5%	51.4	61
Band 6	83	87.5%	90.4%	72.6	75
Band 7	83	98.2%	100.0%	81.5	83
Band 8	84	99.9%	100.0%	83.9	84

Diagnostic	Estimate	Interpretation
Calibration intercept	0.165	Near 0 means predicted risk is not systematically high or low
Calibration slope	1.025	Near 1 means predicted probabilities are not overly extreme or compressed

Subgroup calibration checks show where monitoring would matter before operational use. The table reports groups with at least 25 holdout records.

Group	Level	N	Pred	Obs	Gap	Cases
course track	ap	211	33.7%	39.3%	5.7%	83
course track	honors	84	33.9%	29.8%	-4.1%	25
course track	regular	368	64.0%	64.7%	0.6%	238
assessment window	beginning-of-year	405	45.4%	49.9%	4.5%	202
assessment window	end-of-year	261	58.4%	55.6%	-2.9%	145
attendance category	at-risk	60	53.7%	63.3%	9.6%	38
attendance category	high	285	49.6%	49.5%	-0.1%	141
attendance category	normal	321	50.7%	52.3%	1.7%	168

A ranked queue is often more useful than a single classification cutoff. The lift chart shows how concentrated support-risk cases are in the highest predicted-risk deciles.

Decile	N	Pred	Obs	Cases	Lift	Capture
1	66	99.9%	100.0%	66	1.92x	19.0%
2	67	99.1%	100.0%	67	1.92x	38.3%
3	66	95.2%	98.5%	65	1.89x	57.1%
4	67	82.0%	86.6%	58	1.66x	73.8%
5	67	58.8%	70.1%	47	1.35x	87.3%
6	66	35.2%	24.2%	16	0.47x	91.9%
7	67	18.8%	17.9%	12	0.34x	95.4%
8	66	10.1%	6.1%	4	0.12x	96.5%
9	67	4.4%	11.9%	8	0.23x	98.8%
10	67	2.0%	6.0%	4	0.11x	100.0%

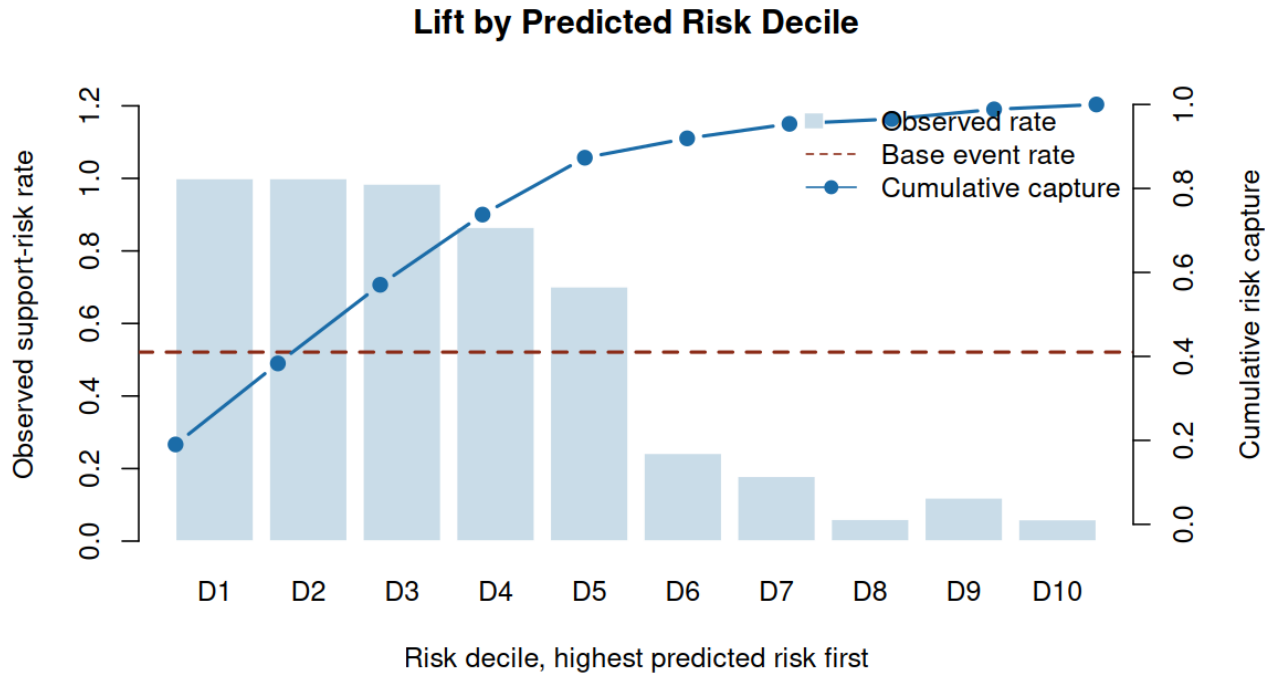


Figure 5: Lift by predicted risk decile

Sensitivity Check

The sensitivity analysis lowers the support-risk score cut point from 50 to 45 and refits the selected model family. This tests whether the prioritization story depends on one particular threshold definition.

Measure	Primary	Sensitivity
Primary holdout event rate	52.1%	Reference
Sensitivity holdout event rate	Reference	41.0%
Sensitivity holdout log loss	0.309	0.329
Sensitivity holdout Brier score	0.093	0.098
Sensitivity holdout AUC	0.938	0.923
Rank correlation with primary predictions	Reference	0.989
Top-quintile overlap with primary ranking	Reference	97.8%
Students changing risk category	Reference	159 of 666

Scenario Profiles

Scenario profiles translate the model into concrete, public-safe support-planning examples with probability intervals.

Scenario	Grade	Track	Window	Attendance	Readiness	Risk	95% CI	Category
On-track check-point	10	regular	beginning-of-year	normal	72.0	2.1%	1.3% to 3.6%	Monitor
Attendance watch	9	regular	beginning-of-year	high	51.0	19.3%	10.1% to 33.7%	Monitor
Priority support	11	honors	end-of-year	at-risk	39.0	99.0%	97.9% to 99.5%	Priority

Bottom Line

- Start with the 50% review threshold as a planning default, then adjust for staffing capacity and support cost.
- Use the ranked queue to prioritize human review and early support planning, not to automate student-level decisions.
- Keep the piecewise readiness model because it captures the discovered nonlinear risk pattern while staying

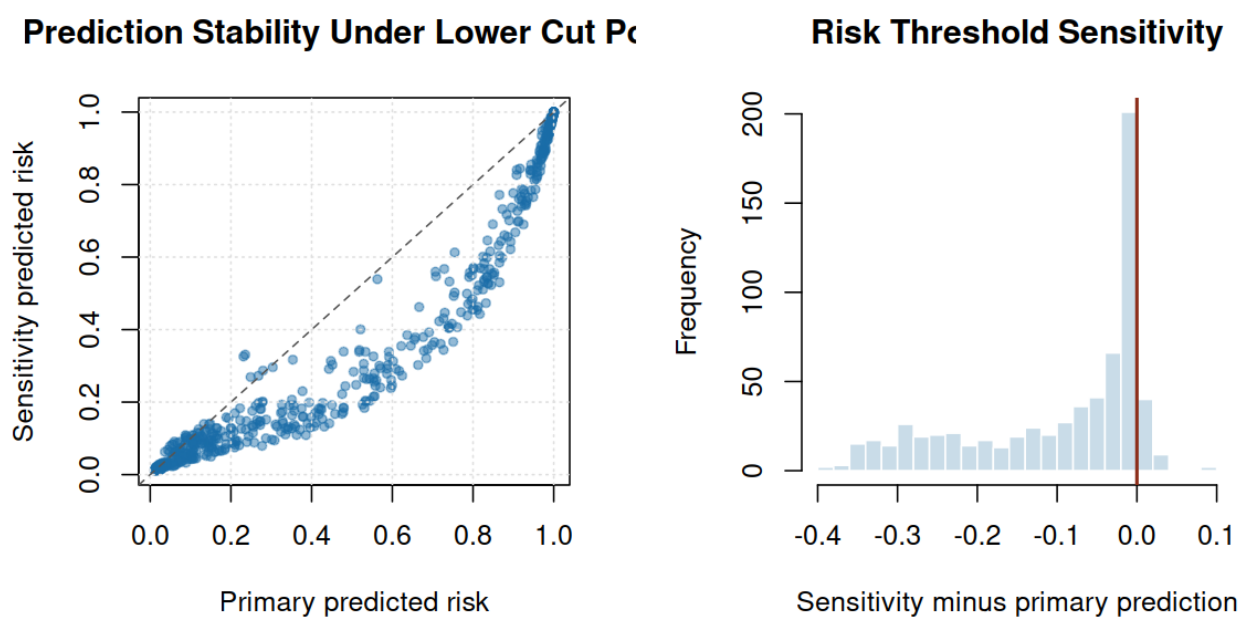


Figure 6: Sensitivity analysis

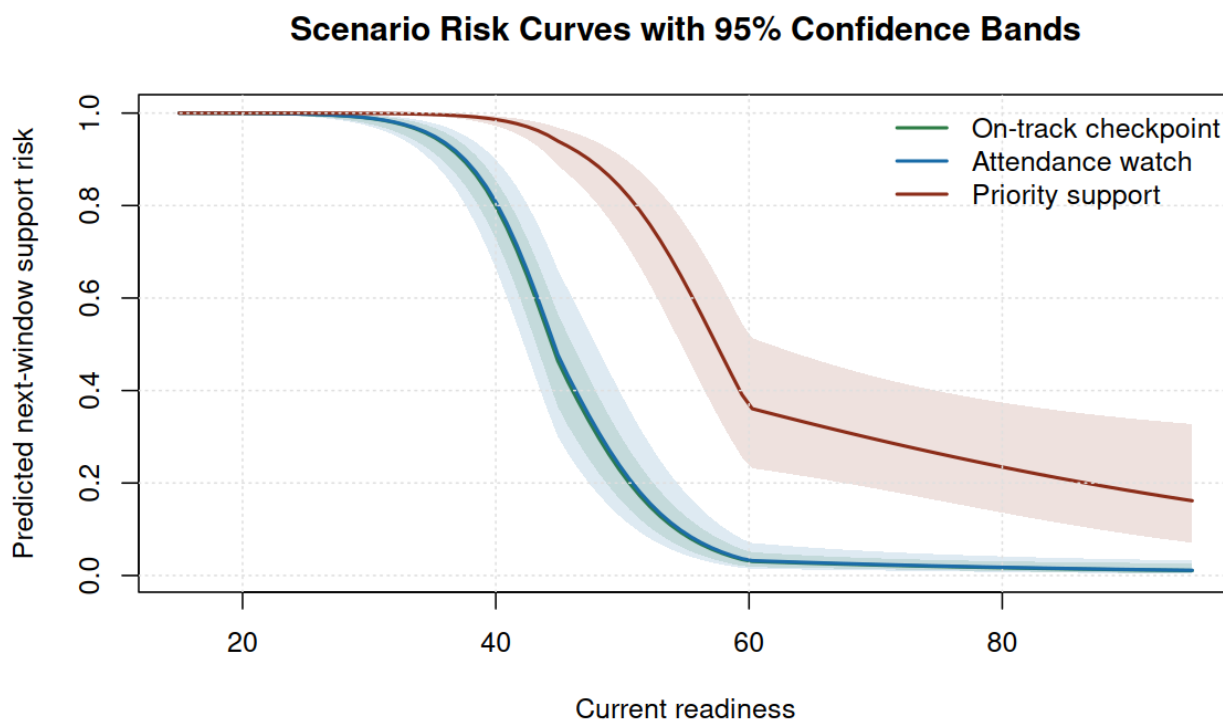


Figure 7: Scenario readiness risk curves